Investigating Class Imbalance Handling Techniques for Machine Learning-Based Network Intrusion Detection

Yuxi He
Postgraduate Cyber Security student
The University of Queensland

Abstract

Class imbalance remains a significant challenge in machine learning-based network intrusion detection, particularly for identifying rare attack categories. This study investigates four imbalance-handling techniques using the UNSW-NB15 dataset: class weighting, random oversampling, random undersampling, and SMOTENC. Logistic Regression and Random Forest are employed as baseline classifiers, with imbalance-handling experiments focusing on Random Forest.

Five-fold Stratified Group Cross-Validation is conducted to reduce overlap between identical feature patterns across training and validation subsets. Class weighting achieves the highest mean Balanced Accuracy (0.6585, compared with 0.5416 for the unweighted Random Forest baseline) and is higher than every other method in all five folds. Its mean Macro-F1 (0.6041, SD 0.0124) is also the highest, but its margin over random undersampling (0.5967, SD 0.0162) is small and within fold-to-fold variation.

On the official test set, class weighting improves Macro-F1 from 0.4585 to 0.5010 and substantially increases recall for minority attack categories, including Backdoor (0.0892 to 0.5763) and Worms. However, these improvements are accompanied by reduced overall accuracy and substantial false positives: Backdoor precision is only 0.0502. The official test set had also been inspected during earlier baseline experiments, which limits how strictly the test results can be read as an untouched evaluation.

Further analysis identifies duplicated feature patterns, conflicting labels, and train-test feature overlap. The findings highlight the importance of balancing minority-class detection performance, precision, and evaluation reliability in network intrusion detection research.

Keywords: Network Intrusion Detection, Class Imbalance, UNSW-NB15, Random Forest, SMOTENC, Machine Learning

1. Introduction

Machine learning-based network intrusion detection systems (NIDS) aim to distinguish malicious from legitimate network traffic. However, substantial class imbalance can make rare attack categories difficult to detect reliably.

The UNSW-NB15 benchmark contains 175,341 training records covering normal traffic and nine attack categories [1]. While the training set includes 56,000 Normal samples, only 130 belong to the Worms category, illustrating the severity of class imbalance.

This study evaluates four imbalance-handling techniques using Random Forest: class weighting, random oversampling, random undersampling, and SMOTENC. Performance is assessed using Macro-F1, Balanced Accuracy, and per-class Precision and Recall, with particular attention to minority-class detection and false-positive trade-offs.

Additionally, duplicated feature patterns, conflicting labels, and train-test feature overlap are investigated to assess potential limitations in the reliability of model evaluation.

2. Methodology

2.1 Dataset, Preprocessing and Models

The experiments use the official UNSW-NB15 training (175,341 records) and test (82,332 records) partitions, with 42 input features: 39 numerical and three categorical (`proto`, `service`, and `state`).

Numerical features undergo median imputation and standardisation; categorical features undergo most-frequent imputation and one-hot encoding. Preprocessing is fitted within training partitions to avoid validation leakage.

A binary Logistic Regression model and multiclass Logistic Regression and Random Forest models are implemented using scikit-learn [3]. Random Forest uses 100 trees and a fixed random seed of 42.

2.2 Imbalance-Handling Techniques

Four Random Forest configurations are compared against an unweighted baseline:

1. Class Weighting: Uses balanced class weights.
2. Random Oversampling: Duplicates minority samples to match the largest class.
3. Random Undersampling: Reduces Normal, Generic, and Exploits to 15,000 samples each.
4. SMOTENC: Increases Analysis, Backdoor, Shellcode, and Worms to 5,000 samples each using a categorical-aware extension of SMOTE [2].

Resampling methods are implemented using imbalanced-learn [4] and applied only to training data within each evaluation split.

2.3 Evaluation Protocol

Five-fold Stratified Group Cross-Validation is performed on the official training partition, grouping records by identical input feature patterns to prevent such patterns from crossing training-validation boundaries.

Macro-F1 and Balanced Accuracy guide model selection, supplemented by Accuracy and per-class Precision, Recall, and F1-score.

The selected class-weighted Random Forest is retrained on the full training partition and evaluated on the official test set. A separate analysis of test records without matching training feature patterns examines sensitivity to test-set composition.

Model selection among the imbalance-handling methods used only cross-validation on the training partition. However, the official test set had already been used to evaluate the Logistic Regression and Random Forest baselines before the imbalance experiments, so it was not completely untouched when the final model was evaluated.

3. Results

3.1 Cross-Validation Performance

Table 1 summarises performance across five grouped cross-validation folds. Fold-level values are printed in notebook 04; the standard deviation (SD) across folds was saved for Macro-F1 only.

Table 1. Five-fold cross-validation results (mean across folds).

Method	Macro-F1 (mean)	Macro-F1 SD	Balanced Accuracy (mean)
Baseline RF	0.5616	0.0130	0.5416
Class Weight	**0.6041**	0.0124	**0.6585**
Random Oversampling	0.5883	0.0096	0.6170
Random Undersampling	0.5967	0.0162	0.6019
SMOTENC	0.5694	0.0086	0.5991

The clearest result is in Balanced Accuracy. Class weighting improved on the baseline by 0.1168 and on the next-best method (random oversampling) by about 0.04, and it had the highest Balanced Accuracy in every one of the five folds.

The Macro-F1 picture is less clear. Class weighting had the highest mean Macro-F1, 0.0425 above the baseline, but only about 0.0075 above random undersampling. That margin is smaller than either method's fold SD, and class weighting had the higher Macro-F1 in only three of the five folds. On an earlier single grouped validation split, random undersampling was slightly ahead (0.6060 vs 0.6034). No statistical significance test was performed, so these results do not show that class weighting is superior on Macro-F1.

3.2 Official Test Set Performance

Table 2. Official test set performance.

Metric	Baseline RF	Weighted RF
Accuracy	0.7544	0.6921
Macro-F1	0.4585	**0.5010**
Balanced Accuracy	0.4828	**0.5871**

The weighted model improved Macro-F1 and Balanced Accuracy at the expense of overall Accuracy. Backdoor recall increased from 0.0892 to 0.5763, Shellcode from 0.6667 to 0.8016, and Worms from 0.0682 to 0.5227. However, Backdoor precision was only 0.0502, indicating substantial false positives.

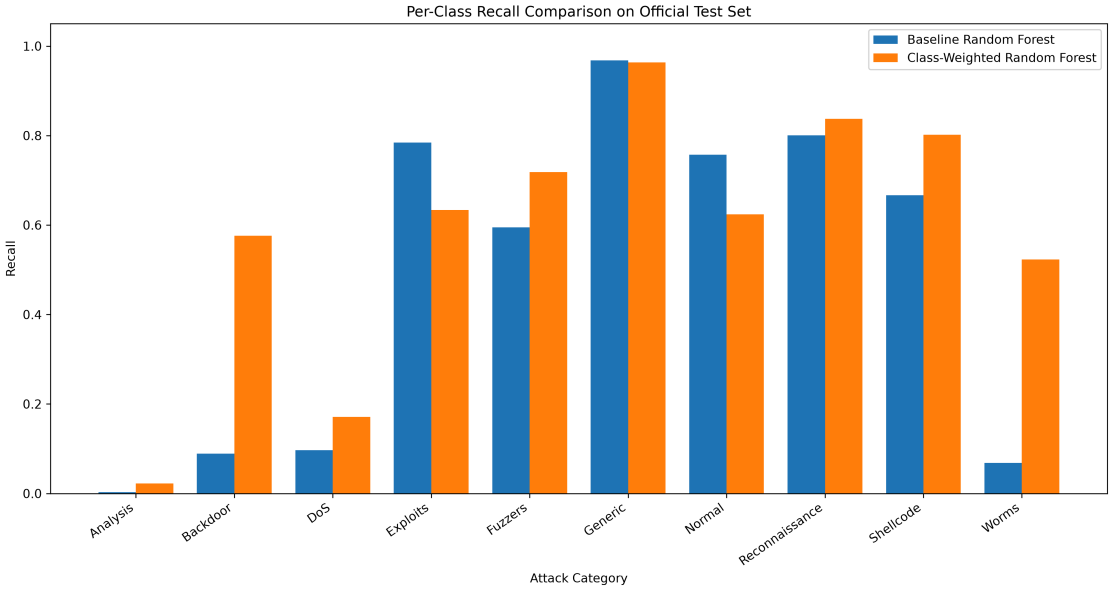


Figure 1. Per-class recall of baseline and class-weighted Random Forest models on the official test set.

3.3 Data Reliability Analysis

Feature-pattern analysis identified 8,541 official test records (10.37%) matching patterns in the training set. On the remaining 73,791 records, the weighted model achieved Macro-F1 of 0.5328 and Balanced Accuracy of 0.6208.

These differences suggest sensitivity to test-set composition but do not establish that overlapping patterns caused the performance changes.

4. Discussion and Limitations

Class weighting gave the clearest cross-validation gain in Balanced Accuracy, while its Macro-F1 advantage over random undersampling was within fold-to-fold variation. It also revealed important trade-offs on the official test set. Although minority-class recall improved, overall Accuracy decreased, and false positives remained substantial. The weighted model produced 6,351 false-positive Backdoor predictions and misclassified 11,442 Normal records as Fuzzers. Backdoor precision was only 0.0502, highlighting the potential burden of false alarms in practical intrusion detection.

Data quality also complicates evaluation. UNSW-NB15 contains duplicated feature patterns and conflicting labels for identical feature inputs. Grouped cross-validation prevents identical patterns from appearing across training-validation boundaries, but cannot eliminate other dependencies. Furthermore, 10.37% of official test records share feature patterns with the training set. Performance differences on the non-overlapping subset may reflect changes in data composition rather than the direct effect of overlap.

Several limitations remain. Experiments were restricted to one dataset and one random seed, with only 44 Worms records in the official test set. Extensive hyperparameter optimisation, external validation, repeated cross-validation, and statistical significance testing were not conducted, so small differences between

methods, such as the Macro-F1 gap between class weighting and random undersampling, cannot be resolved. Additionally, the official test set had been used to evaluate the baseline models during earlier experiments and was therefore not completely untouched before final model evaluation.

Consequently, the results support class weighting for improving Balanced Accuracy under the evaluated conditions, but they do not establish that it is statistically or universally superior. Future work should investigate external datasets and methods that improve minority-class recall while controlling false positives.

5. Conclusion

This study compared four class imbalance-handling techniques for Random Forest-based network intrusion detection using UNSW-NB15. Under five-fold grouped cross-validation, class weighting achieved the highest mean Balanced Accuracy (0.6585), consistently across folds. Its highest mean Macro-F1 (0.6041) was not clearly separated from random undersampling (0.5967).

On the official test set, class weighting improved Macro-F1 from 0.4585 to 0.5010 and Balanced Accuracy from 0.4828 to 0.5871, with substantial recall gains for rare attacks such as Backdoor and Worms. However, overall accuracy declined, and false positives remained problematic.

The findings demonstrate that improved minority-class detection must be evaluated alongside precision and data quality. Future work should examine external datasets and approaches that balance minority-class sensitivity with false-positive control.

6. References

[1] Moustafa, N., & Slay, J. (2015). UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). 2015 Military Communications and Information Systems Conference (MilCIS), 1-6. https://doi.org/10.1109/MilCIS.2015.7348942

[2] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321-357. https://doi.org/10.1613/jair.953

[3] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830. https://jmlr.org/papers/v12/pedregosa11a.html

[4] Lemaître, G., Nogueira, F., & Aridas, C. K. (2017). Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning. Journal of Machine Learning Research, 18(17), 1–5. https://jmlr.org/papers/v18/16-365.html